BUS 9430 · CAPSTONE 2 · DATA UNDERSTANDING AND PREPARATION

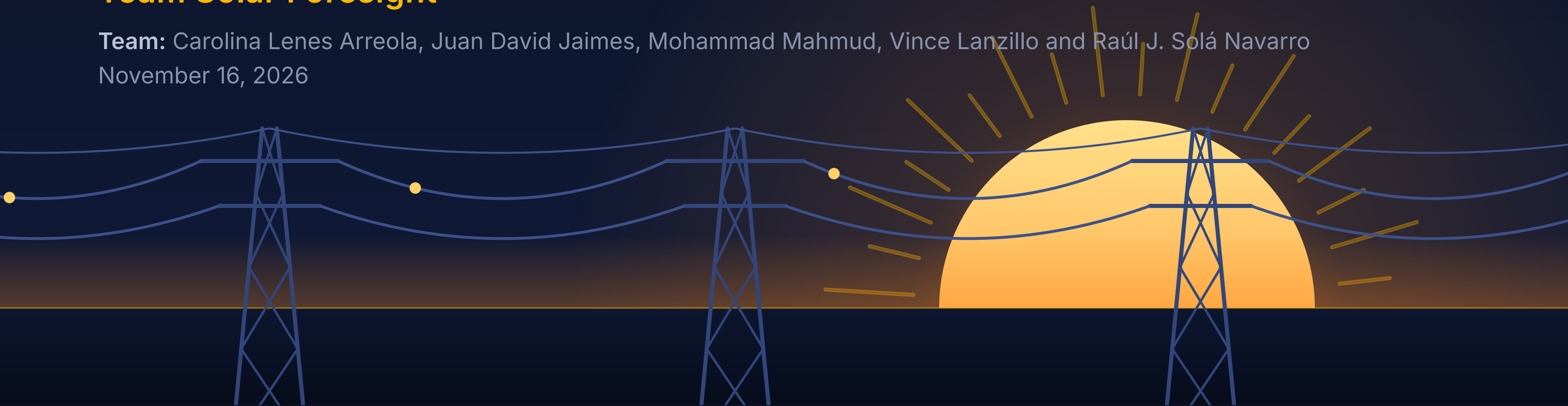
Tomorrow's Sun, Today's Plan

What our data shows, and how we get it ready

Team Solar Foresight

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THE BUSINESS PROBLEM

Solar is the part of tomorrow's plan that depends on the weather

WHAT CAISO DOES

Locks in tomorrow's plan today

Bids close at 10 a.m. Pacific. The plan is out by 1 p.m.

WHAT GOES WRONG

The solar forecast misses

Too high: backup power bought at the last minute. Too low: reserves held for nothing.

WHAT WE FORECAST

Hourly solar output, a day ahead

In megawatts (MW), using only what is known at 10 a.m. Not demand: CAISO already forecasts that.

THE DATA

Five public sources, one hourly table

ANSWER KEY
EIA
Actual solar output, hour by hour

INPUTS
Open-Meteo
Weather forecasts made before 10 a.m.

BENCHMARK
CAISO
Its own day-ahead solar forecast

SCALE
EIA Form 860M
Installed solar capacity

FLAGS
CAISO curtailment
Solar that was turned down

PERIOD
Mar 2024 to Oct 2026
954 days

HOURLY ROWS
22,896
11,733 in daylight

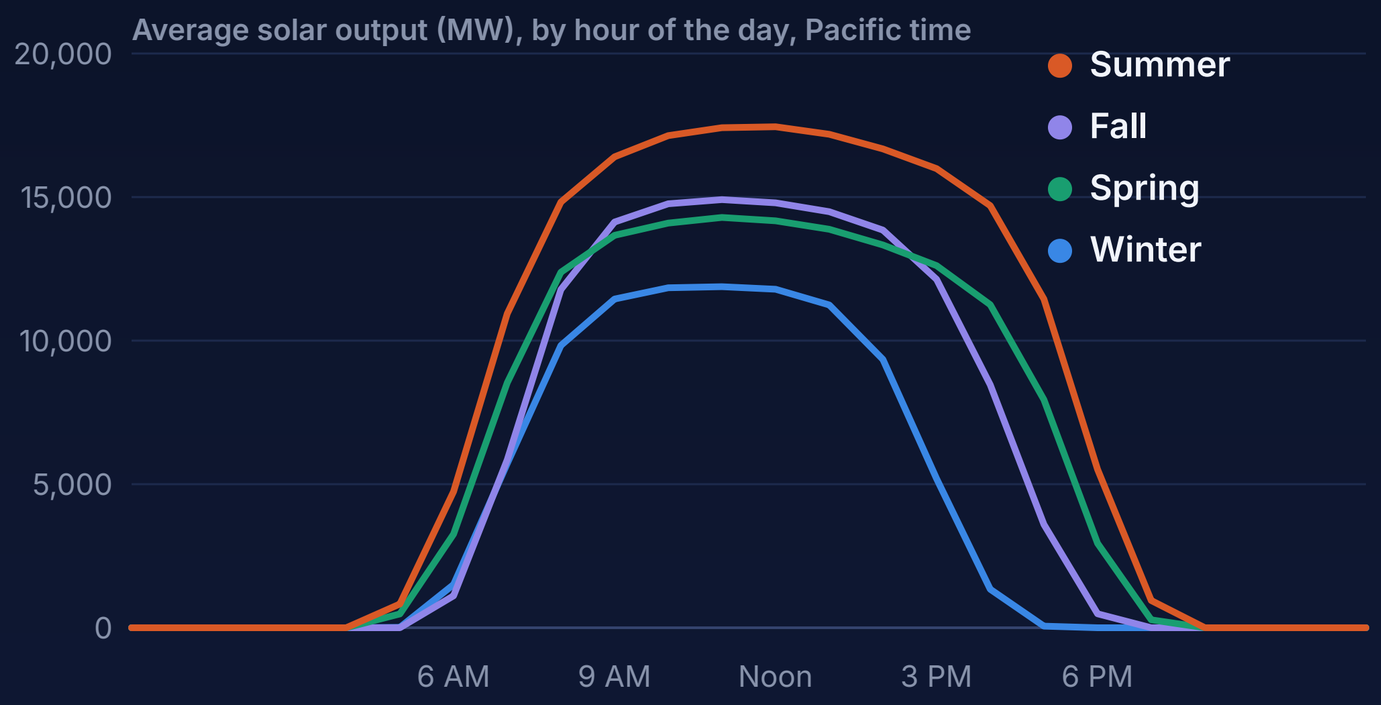
HOURS WITH A GAP
< 1%
225 hours, all documented

Privacy: low. Public, system-wide data, no personal information. **Delivery:** raw files saved in our repository; the weather service's free tier is non-commercial and capped per day.

Weather data by Open-Meteo.com (CC BY 4.0).

WHAT THE DATA SHOWS

A regular daily shape, driven by season and weather



+47%

Summer midday peak versus winter.

68% vs 41%

Midday output on clear versus very cloudy forecast days, as a share of capacity.

+24%

Installed solar since March 2024, and up to 1 in 4 possible MWh turned down in spring.

Average solar output by hour, March 2024 to April 2026, Pacific time. Sources: EIA, Open-Meteo, CAISO.

GETTING IT READY

From raw data to a fair test

1. **One clock**

   Every source on the same hour.

2. **Fix gaps**

   Fill short ones, drop long ones.

3. **Flag turned-down solar**

   Report with and without.

4. **Scale by capacity**

   So months compare fairly.

Mar 2024 — Oct 2026

Teach and tune the models: earliest 80% of days (four rolling checks) | **Locked test: 20%**

The test is the latest 191 days (April 10 to October 17, 2026). It stays locked until the end and has no winter, so we judge winter in validation.

THE BAR TO BEAT

Our scorecard starts with CAISO

SAME AS YESTERDAY

4.9%

Average miss, as a share of capacity. The floor to beat.

CAISO, AS PUBLISHED

5.7%

About 830 MW too high on average.

CAISO, ADJUSTED

3.7%

With turned-down solar added back. The fairest picture of CAISO's skill.

OUR TARGET: NO MORE THAN 1.25 TIMES CAISO'S MISS

4.6% to about 7%, depending on which CAISO number counts

WATCHING

Short history, no winter in the test, weather timing at 10 a.m.

NEXT

Compare models Nov 17 to Dec 8. Results Dec 14.